

This section evaluates the potential environmental impacts of the proposed project related to hydrology and water quality. The existing surface water and groundwater hydrologic conditions of the proposed project and in the surrounding area are characterized, and a summary is provided of relevant laws and regulations as they apply to the proposed project. The impact analysis focuses on potential issues associated with increased stormwater runoff and water quality. Draft EIR Section 3.14, Utilities and Service Systems, evaluates impacts related to water supply.

Public comments received on the Notice of Preparation (NOP), included in **Appendix 1.0** of this Draft EIR, requested evaluation of project impacts on surface water, groundwater, and water supply. As noted above, water supply is evaluated in Section 3.14. The Contra Costa County Flood Control and Water Conservation District requested information about watersheds, drainage facilities, sources of stormwater runoff, and how the City would manage stormwater discharges for the proposed project. The comment letter also summarized laws and permit requirements. The analysis in this section has considered those comments.

3.8.1 EXISTING SETTING

CLIMATE AND TOPOGRAPHY

Orinda is in a part of the San Francisco Bay Area characterized by a dry-summer subtropical climate (often referred to as Mediterranean), with cool wet winters and relatively warm, dry summers. The topography in Orinda varies from low-lying valleys to rolling hills. The crests of the hills range in elevation from 1,000 to 1,100 feet.

SURFACE WATER RESOURCES

Most of Orinda is within the San Pablo Creek watershed. A small area in the eastern part of the City is in the Las Trampas Creek watershed, and the southwest corner of the City is in the San Leandro Creek watershed. Watershed boundaries are shown in **Figure 3.8-1**.

The San Pablo Creek watershed covers approximately 43.5 square miles in western Contra Costa County. In addition to Orinda, this watershed includes San Pablo and Richmond as well as parts of unincorporated Contra Costa County. There are numerous unnamed creeks, streams, and tributaries, and numerous subwatersheds, in the San Pablo Creek watershed. Major water bodies in this watershed include San Pablo Creek, Cascade Creek, Lauterwasser Creek, Bear Creek, Castro Creek, Siesta Valley Creek, Wilkie Creek, Lake Cascade, San Pablo Reservoir, and Briones Reservoir. The San Pablo Creek watershed supports 10,909 acres of East Bay Municipal Utility District (EBMUD)–protected watershed land, including the Siesta Valley, Gateway, and Briones watersheds. Impervious surfaces make up approximately 20 percent of the watershed (EBMUD 2006).

San Pablo Creek, the main surface water feature in the City, originates in Orinda at Moraga Creek, flows northwest along the eastern edge of the Oakland Hills to San Pablo Reservoir, and ultimately discharges to San Francisco Bay near Richmond. San Pablo Creek has over 30 named tributaries and numerous additional unnamed tributaries (FEMA 2013). Lauterwasser Creek, a perennial tributary, flows into San Pablo Creek approximately one-third mile southeast and upstream of the Orinda water treatment plant, and upstream from the confluence of San Pablo Creek and San Pablo Reservoir (EBMUD 2006). **Figure 3.8-1** shows the locations of major surface water features in Orinda.

3.8 HYDROLOGY AND WATER QUALITY

Orinda's creeks contribute to the semi-rural character of the City and provide valuable riparian habitat. San Pablo Creek is channelized in the downtown area, and Lauterwasser Creek is underground at Dalewood Drive, but creeks in other locations are in a natural state. The Contra Costa County Flood Control and Water Conservation District (CCCFCWCD) is responsible for maintenance of the improved sections of San Pablo Creek, but maintenance of creeks elsewhere is the responsibility of landowners (Orinda 1987). Orinda is located in CCCFCWCD Drainage Area 53 (CCCFCWCD 2014).

STORM DRAINAGE

Stormwater in Orinda is conveyed via a combination of privately owned storm drain improvements and a storm drainage system managed by the City's Public Works and Engineering Services Department. The City's storm drainage system has been in service for over 50 years, and some sections of the drainage facilities are currently in need of major repair and, in some instances, rebuilding. Budgetary constraints require the City to carefully select the existing drainage facilities that need to be repaired or replaced. The Orinda Citizens' Infrastructure Oversight Commission adopted selection criteria in 2012 for selection of drainage improvements, with the intent of assisting City staff with identifying and selecting projects to be recommended in the City's Annual Drainage Improvement Program. Pipes or drainage structures that have failed typically are high priority.

Recently funded projects in the City's recently adopted Capital Improvement Program (CIP) include emergency repairs to the Tarabrook Drive storm drain, North Lane Storm Water Improvement Project, and Oak Springs/Candlestick Road stormwater improvements, along with drainage improvements identified in conjunction with funded roadway improvements. The CIP also included funding for a drainage master plan (currently being prepared by the City's consultant) that will help identify future drainage repairs to be added to the CIP (Orinda 2014). When completed, the North Lane will provide greater storm flooding hazard protection by increasing the capacity of the line and creating an outfall structure at San Pablo Creek north of Downtown along Camino Pablo at North Lane near the EMBUD Orinda Water Treatment Plant (FEMA 2013).

City staff observations and preliminary results of the drainage master plan study indicate storm drain capacity is generally not a problem in Orinda. This is primarily because of the City's hilly terrain; stormwater tends to flow quickly downstream and out of the City. Capacity issues are generally related to the age and condition of the storm drain lines themselves (as noted above) and not the rate or volume of flow from development. The City ensures storm drain capacity is sufficient to accommodate new flows in addition to existing flows by requiring development projects to include appropriate on-site controls and designing inlets and pipe sizing in accordance with City standards (Chapter 16.64 of the Municipal Code; see Section 3.8.2, Regulatory Framework, below)(Wellbrock 2014).

The City does not allow stormwater generated by projects in the City to be discharged into any water conveyance facility of the Contra Costa Canal (Municipal Code Section 16.64.050).

GROUNDWATER RESOURCES

Orinda is underlain by sedimentary and volcanic bedrock with some areas of alluvium along stream channels (see Section 3.5, Geology and Soils). Groundwater depth in the City is assumed to be shallow, perched groundwater in localized areas. The California Department of Water Resources (DWR) (2003) has not delineated any groundwater basins in the Orinda. Groundwater recharge is generally limited to areas along creeks and streams, where alluvial soils are present. Groundwater in Contra Costa County is not a source of EBMUD water supply (EBMUD 2011).

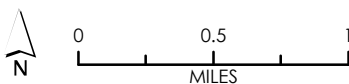
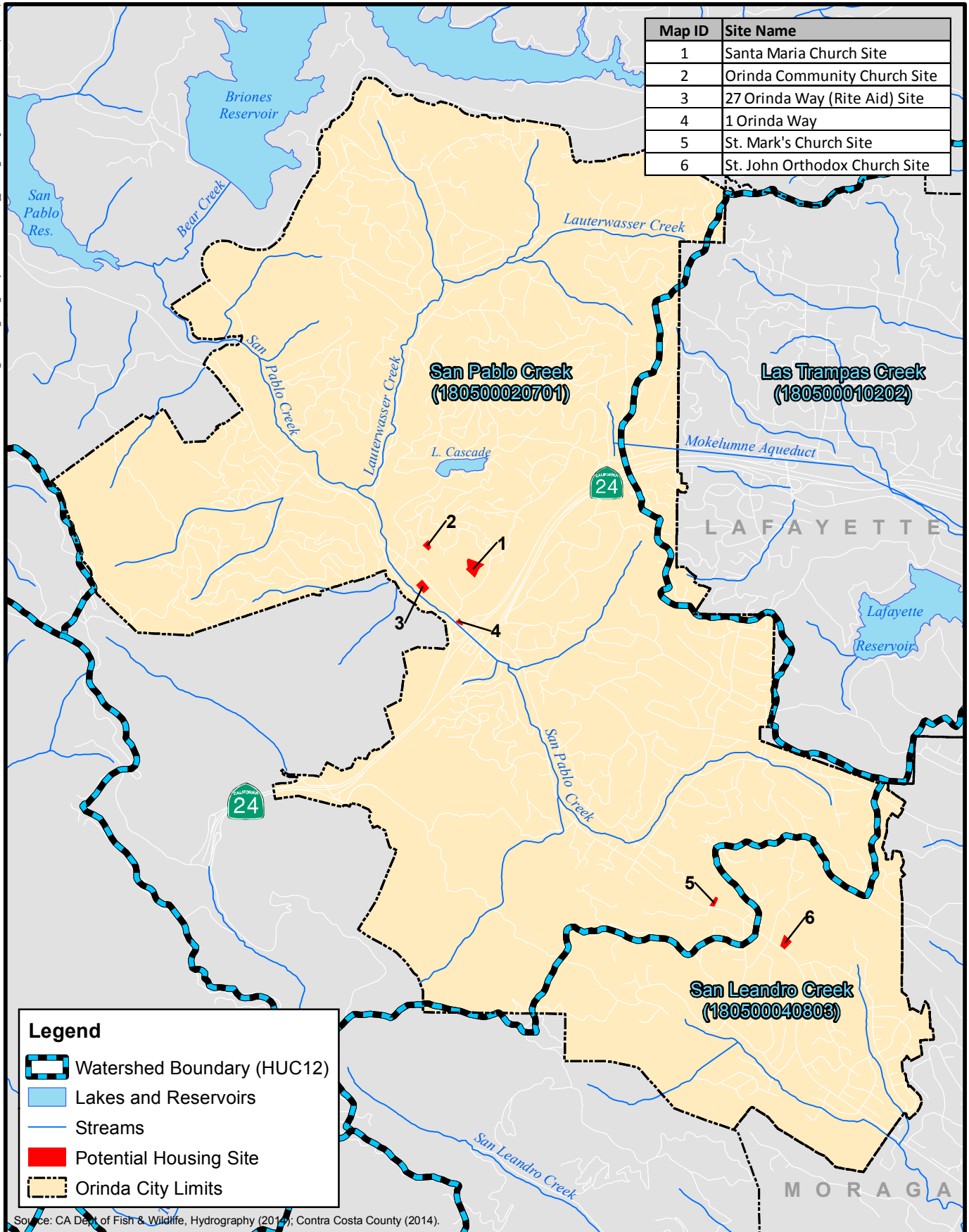


Figure 3.8-1
Watersheds

WATER QUALITY

Water quality monitoring on San Pablo Creek in 2013 indicated water temperature, dissolved oxygen, and pH values all met the water quality criteria established in the San Francisco Bay Basin Water Quality Control Plan (CCCWP 2014).¹ San Pablo Creek is an impaired water body for diazinon from urban runoff under Section 303(d) of the federal Clean Water Act, and the US Environmental Protection Agency approved a TMDL for diazinon in 2007.² A TMDL for trash is required by 2021 (SWRCB 2011).

The City of Orinda implements a comprehensive stormwater runoff water quality protection program, as required by federal and state laws and regulations (Chapter 18.03 of the Municipal Code; see subsection 3.8.2, Regulatory Framework, below). The City performs field inspections (storm drain system, commercial businesses, construction sites, and illicit connections), conducts education outreach activities (creek cleanups, K–12 education, storm drain stenciling), implements Integrated Pest Management techniques, performs water quality monitoring, and uses full trash capture devices.

The San Pablo and Briones reservoirs, which are part of the EBMUD water supply system, are just north of the City. In order to minimize the risk of reservoir pollution from urban development, EBMUD has acquired much of the surrounding watershed, including areas in Orinda. EBMUD (1999) has adopted the East Bay Watershed Master Plan that contains policies to maintain and enhance water quality. City of Orinda General Plan policies support the concept of keeping much of the land on the north and west of the City that drains into the EBMUD reservoirs designated as parks, very low-density residential, and watershed land to protect water quality and aesthetic values.

FLOOD HAZARDS

Dams

Three dams regulated by the State of California are location in the Orinda area: San Pablo Dam and Briones Reservoir, owned and operated by EBMUD, and Lake Cascade Dam, owned and operated by the Orinda Country Club. The DWR has broad responsibilities over these dams to ensure they meet safety standards. No dam in Contra Costa County has failed as a result of earthquake ground shaking.

There are 105 acres at risk of flood hazards from dam failure inundation in Orinda (Orinda 2011). Failure of San Pablo Dam would not cause flooding in Orinda because floodwaters would flow toward San Francisco Bay. Inundation maps developed by Contra Costa County show that in the event of dam failure at Briones Reservoir, land south of Briones Dam to San Pablo Reservoir would flood, as would land immediately east of San Pablo Dam Road and Camino Pablo. The Briones Dam inundation area is in an area zoned for low-density residential uses. If Lake Cascade Dam were to fail, flooding would extend almost to Downtown Orinda (Orinda 1987). However, the mapped inundation area is zoned for nonresidential uses.

¹ The San Francisco Bay Regional Water Quality Control Board (RWQCB) regulates water quality in San Francisco Bay under the Porter-Cologne Water Quality Control Act through regulatory standards and objectives in the Water Quality Control Plan for the San Francisco Bay Basin (Basin Plan). The Basin Plan identifies existing and potential beneficial uses and provides numerical and narrative water quality objectives to protect those uses.

² Section 303(d) establishes the total maximum daily load (TMDL) process, which requires states to identify waters whose water quality is “impaired” (affected by the presence of pollutants or contaminants), and to establish a TMDL or the maximum quantity of a particular contaminant that a water body can assimilate without experiencing adverse effects on the beneficial use identified.

100-Year Flood Hazard Areas

The Federal Emergency Management Agency (FEMA) has delineated flood hazard areas in Orinda. The 100-year floodplain is confined to a few streams and creeks, including San Pablo Creek through the downtown area, and Lake Cascade (FEMA 2009). Approximately 117 acres within the City limits—or about 1.4 percent of the City's area—is within the 100-year floodplain (Zone A). Most of the flood-prone properties are located along San Pablo Creek (Orinda 2011). Flooding, in general, does not present a significant risk to Orinda, although there is the potential for local flood damage caused by overtopping creeks during storms (Orinda 1987).

Seiche, Tsunami, and Mudflow

Orinda is not at risk of tsunami due to its inland location. Based on mapping prepared by the US Geological Survey (USGS) (1997), the east-facing slopes on the west side of the City have a high concentration of debris flow source areas, which includes mudflow. Much of the west side is undeveloped EBMUD watershed land and open space; however, there is moderately dense residential development northwest of Downtown (Claremont and El Toyonal areas), which may be an area susceptible to mudflow because it is downslope from debris flow source areas. There are other scattered locations throughout the City where steeper slopes are present that are also debris flow source areas. Those areas also contain numerous residential properties, but there are fewer potential mudflow source areas.

Lake Cascade (also referred to as Cascade Lake and Lake Orinda) is a small impoundment in the Orinda Country Club. There is a potential for seiche in the event of an earthquake; however, there are no occupied land uses surrounding the lake, and potential housing sites would not be located in the country club. The City has not identified seiche as a significant hazard in Orinda (Orinda 2011).

ALTERNATIVE SITES HYDROLOGY AND WATER QUALITY CHARACTERISTICS

The Santa Maria Church (Alternative 1), Orinda Community Church (Alternatives 2 and 3), St. Mark's Church (Alternative 2), 27 Orinda Way (Rite Aid) site (Alternative 3), and 1 Orinda Way (Alternative 3) are in the San Pablo Creek watershed. The St. John Orthodox Church site (Alternative 2) is in the San Leandro Creek watershed.

None of the alternative sites contain any natural creeks or streams, nor are they situated near any creeks or streams. Stormwater runoff from each of the sites is conveyed to the City's storm drain system, which does not have any major capacity problems. Constituents in stormwater runoff from the sites containing paved areas include typical urban pollutants such as oil and grease, metals, sediment, and trash. Where paved areas adjoin vacant grassland or landscaped areas, additional pollutants include bacteria, nutrients, and pesticides.

Because the sites are small, not near any creeks or streams, and contain some impervious surfaces or other features that tend to limit infiltration from precipitation, none of the sites is a significant source of groundwater recharge. There are no groundwater wells on the alternative sites.

None of the alternative sites is in a FEMA 100-year floodplain. Although the Alternative 3 27 Orinda Way (Rite Aid) site is just east of San Pablo Creek, according to FEMA mapping, it is not within the 100-year floodplain or any other flood hazard area (FEMA 2009). The alternative sites are not located in an area that has been mapped by Contra Costa County as being at risk of dam failure inundation. Based on USGS (1997) mapping, the alternative sites do not appear to be in areas susceptible to mudflow hazard. Although gentle slopes adjoin the Santa Maria and

Orinda Community church locations, there is residential development upslope and no vacant or undeveloped land that could be in a potential path for mudflow.

3.8.2 REGULATORY FRAMEWORK

FEDERAL

Clean Water Act

The federal Clean Water Act (CWA) gives states the primary responsibility for protecting and restoring water quality. In California, the State Water Resources Control Board (SWRCB) and the nine Regional Water Quality Control Boards (RWQCBs) are the agencies with the primary responsibility for implementing federal CWA requirements, including developing and implementing programs to achieve water quality standards. Water quality standards include designated beneficial uses of water bodies, criteria or objectives (numeric or narrative) which are protective of those beneficial uses, and policies to limit the degradation of water bodies. The proposed project is located in a portion of the state that is regulated by the San Francisco Bay RWQCB.

Sections 401 and 404 of the CWA

Sections 401 and 404 of the CWA are administered through the regulatory program of the US Army Corps of Engineers (USACE) and regulate the water quality of all discharges of fill or dredged material into waters of the United States, including wetlands and intermittent stream channels. Section 401, Title 33, Section 1341 of the Clean Water Act sets forth water quality certification requirements for any applicant applying for a federal license or permit to conduct any activity including, but not limited to, the construction or operation of facilities which may result in any discharge into the navigable waters.

National Pollutant Discharge Elimination System

As authorized by Section 402(p) of the CWA, the National Pollutant Discharge Elimination System (NPDES) Permit Program controls water pollution by regulating point sources that discharge pollutants into waters of the United States. The SWRCB issues NPDES permits to cities and counties through the Regional Water Quality Control Boards, and it is the responsibility of the RWQCBs to preserve and enhance the quality of the state's waters through the development of water quality control plans and the issuance of waste discharge requirements. Waste discharge requirements for discharges to surface waters also serve as NPDES permits.

National Flood Insurance Program

The Federal Emergency Management Agency oversees floodplains and administers the National Flood Insurance Program (NFIP). Areas of special flood hazard (those subject to inundation by a 100-year flood) are identified by FEMA through regulatory flood maps (Flood Insurance Rate Maps). The NFIP mandates that development cannot occur within the regulatory floodplain (typically the 100-year floodplain) if that development results in an increase of more than 1 foot elevation. In addition, development is not allowed in delineated floodways within the regulatory floodplain.

STATE

Porter-Cologne Water Quality Control Act

The Porter-Cologne Water Quality Control Act established the SWRCB and the nine RWQCBs as the principal state agencies with the responsibility for controlling water quality in California. The act contains water quality policies and water quality standards that apply to both surface water and groundwater, and it provides the legislative framework for regulations governing the discharges of pollutants from point and nonpoint sources.

Water Quality Control Plan for the San Francisco Bay Basin (Basin Plan)

The San Francisco Bay RWQCB regulates water quality in San Francisco Bay under the Porter-Cologne Water Quality Control Act through regulatory standards and objectives in the Water Quality Control Plan for the San Francisco Bay Basin (Basin Plan). The Basin Plan identifies existing and potential beneficial uses and provides numerical and narrative water quality objectives to protect those uses.

General Construction Activity Storm Water Permits and Stormwater Pollution Prevention Plans

The SWRCB has adopted a General Permit for Stormwater Discharges Associated with Construction and Land Disturbance Activities (Construction General Permit) (CAS000002, Waste Discharge Requirements, Order No. 2009-0009-DWQ, as amended by Order No. 2010-0014-DWQ and Order 2012-0006-DWQ). The Construction General Permit applies to any construction activity affecting 1 acre or more. The focus of the permit is to minimize the potential effects of construction runoff on receiving water quality. The permit requires preparation of a stormwater pollution prevention plan (SWPPP) that identifies best management practices (BMPs) describing erosion control measures. Examples of typical construction BMPs to address water quality include using temporary mulching, seeding, or other suitable stabilization measures to protect uncovered soils.

Project proponents are required to submit a Notice of Intent, a site map, a signed certification statement, an annual fee, and a SWPPP. The permit program is risk-based, wherein a project's risk is based on the project's potential to cause sedimentation and the risk of such sedimentation on the receiving waters. A project's risk determines its water quality control requirements, ranging from Risk Level 1, which consists of only narrative effluent standards, implementation of best management practices, and visual monitoring, to Risk Level 3, which consists of numeric effluent limitations, additional sediment control measures, and receiving water monitoring. Additional requirements include compliance with post-construction standards focusing on low impact development (LID), preparation of rain event action plans, increased reporting requirements, and specific certification requirements for certain project personnel.

The SWPPP must include best management practices to reduce construction effects on receiving water quality by implementing erosion control measures and reducing or eliminating non-stormwater discharges. Examples of typical construction best management practices included in SWPPPs include, but are not limited to, using temporary mulching, seeding, or other suitable stabilization measures to protect uncovered soils; storing materials and equipment to ensure that spills or leaks cannot enter the storm drain system or surface water; developing and implementing a spill prevention and cleanup plan; and installing sediment control devices such as gravel bags, inlet filters, fiber rolls, or silt fences to reduce or eliminate sediment and other pollutants from discharging to the drainage system or receiving waters.

Municipal Regional Permit for Stormwater Runoff

The municipalities of the Contra Costa Clean Water Program, the Contra Costa County Flood Control and Water Conservation District, and Contra Costa County and its 19 incorporated cities (including the City of Orinda) are regulated waste dischargers under a Municipal Regional Stormwater NPDES Permit (Order R2-2009-0074, as amended by Order R2-2011-0083; NPDES Permit No. CAS612008) administered by the San Francisco Bay RWQCB. Provision C.3 of the Municipal Regional Permit (MRP) for New Development and Redevelopment allows permittees, such as the City of Orinda, to use their planning authorities to include appropriate source control, site design, and stormwater treatment measures in new development and redevelopment projects to address both soluble and insoluble stormwater runoff pollutant discharges and prevent increases in runoff flows from new development and redevelopment projects. The goal is to be accomplished primarily through the implementation of low impact development techniques (MRP Provision C.3.c). The City of Orinda implements the Provision C.3 requirements through Municipal Code Chapter 18.02.

The MRP defines which projects must comply with the design standards. New and redevelopment projects that create or replace 10,000 square feet or more of impervious surface are subject to MRP Provision C.3. Those projects must provide permanent/post-construction treatment controls for stormwater according to specific calculations. If the redevelopment results in an alteration of more than 50 percent of the existing impervious surfaces, permanent BMPs must be implemented to treat runoff from the entire project site. The CCCWP has developed a Stormwater C.3 Guidebook that identifies the requirements and methods to achieve compliance (CCCWP 2012).

Low Impact Design (LID)

The goal of LID is to reduce runoff and mimic a site's predevelopment hydrology by minimizing disturbed areas and impervious cover, and then infiltrating, storing, detaining, and/or biotreating stormwater close to the source. LID uses principles such as preserving and re-creating natural landscape features and minimizing imperviousness to create a functional and appealing site drainage that treats stormwater as a resource rather than a waste product. MRP Provision C.3.c requires source control and landscaping that minimizes irrigation and runoff and promotes surface infiltration. A regulated project must implement at least one of the design strategies identified in the MRP (e.g., minimizing impervious surfaces and/or directing roof runoff into cisterns). Each regulated project must identify how much stormwater must be treated, and the project is required to treat 100 percent of the amount of that runoff (e.g., using infiltration or biotreatment techniques).

LOCAL

City of Orinda General Plan

Conservation Element

The Conservation Element contains the following guiding policies that are applicable to drainage and water quality:

- 4.1.1.E Protect creeks from siltation, pollution, and debris buildup to minimize the danger of flooding storms, to retain the aesthetic and habitat values of the creeks in their natural state, and enhance and restore them where possible. Prohibit major channelization.

3.8 HYDROLOGY AND WATER QUALITY

The corresponding implementing policies are:

- 4.1.2.E Preserve drainage easements along creeks in order to protect adjacent buildings from flooding, and to preserve valuable riparian vegetation.
- 4.1.2.H Review development proposals to ensure site design and construction methods that minimize soil erosion and volume and velocity of surface runoff, and mitigate impacts on properties below.
- 4.1.2.I Limit development in the proximity of reservoirs to prevent siltation and water contamination.

Safety Element

The Safety Element contains the following guiding policy that is applicable to flood hazards:

- 4.2.1.C Development shall be located away from flood-prone areas unless flood risks can be mitigated.

The corresponding implementing policy is:

- 4.2.2.E Land development shall be consistent with the natural carrying capacity of nearby creeks, streams and other waterways.

Growth Management Element

The Growth Management Element establishes performance standards for services to ensure, generally, that public facilities are provided consistent with the standards adopted in the element. Performance standard 5.4.2.F requires enforcement of existing ordinances regulating development in floodplains (Ordinance 04-08). Performance standard 5.4.3.D requires that all new development projects contribute to or participate in the improvement of flood control systems in proportion to the demand generated by project occupants and users as determined by the City.

City of Orinda Municipal Code

Chapter 16.64 (Storm Drainage)

Section 16.64.020 of the Municipal Code requires that the design of storm drain improvements must be such that all surface waters occurring generated by a project³ regulated by this code, as well as all surface waters flowing onto and/or through the project, must be conveyed through without damage to any improvement, building site, or dwelling. Storm drainage facilities within must be designed to adequately convey the stormwater runoff from the ultimate development of the drainage basin or watershed. The rate and volume of stormwater from a development project must be determined in accordance with Section 16.64.060 of the Municipal Code. Section 16.64.070 identifies capacity criteria for designing project drainage infrastructure. The storm drainage plan must be submitted to the City for review and approval prior to the issuance of a grading permit.

³ Chapter 16.04 development requirements for subdivisions. Section 16.08 defines subdivision to include a community apartment project as defined in subdivision (d) of Section 1351 of the Civil Code.

Chapter 18.02 (Stormwater Management and Discharge Control)

Municipal Code Chapter 18.02 implements the MRP Provision C.3 requirements. Section 18.02.040.B requires that every application for a development project, including but not limited to a rezoning, tentative map, parcel map, conditional use permit, variance, site development permit, design review, or building permit that is subject to the development runoff requirements in the MRP, be accompanied by a stormwater control plan that meets the criteria in the most recent version of the Contra Costa Clean Water Program Stormwater C.3 Guidebook (CCCWP 2012). Under Section 18.02.040.B, implementation of an approved stormwater control plan and submittal of an approved stormwater control operation and maintenance plan by the applicant must occur prior to issuance of a certificate of occupancy for a project subject to the requirements of Chapter 18.02. Section 18.02.040.C requires that all stormwater management facilities be designed in a manner to minimize the need for maintenance and reduce the chances of failure. To comply with Section 18.02.040.D, all stormwater management facilities must be maintained according to the Guidebook and the approved stormwater control operation and maintenance plan. The stormwater operation and maintenance plan must describe how the maintenance costs will be funded. Those responsible for maintenance must inspect the stormwater management facility at least annually. These requirements would apply to the construction of multi-family housing units at the alternative sites.

Chapter 18.03 (Watercourse Maintenance, Alteration, and Protection)

Projects that result in a net increase of 500 square feet or more of impervious surface area must have a drainage permit (Municipal Code Section 18.03.050). The City requires that all permit applicants with projects that exceed 500 square feet of new impervious surfaces or that alter existing drainage systems route their runoff to landscaped areas and away from the street and creeks to the extent practicable. Drainage plans must be submitted to the City for review and approval. Projects that are subject to a drainage permit include new homes or additions of 1,500 square feet or more; new structures or additions to existing structures that result in an expansion of the building footprint by at least 500 square feet; excavating or grading projects subject to a grading permit (if grading is more than 50 cubic yards, an erosion control plan is required); and installing, altering, or expanding a stormwater drainage system. All projects creating or replacing 2,500 square feet or more of impervious surface must submit and obtain City approval of a stormwater control plan that meets specific criteria.

In addition, under Section 18.03.050, no person, firm, or corporation may impair, impede, redirect, or affect waters in a watercourse;⁴ construct, alter, or repair a watercourse; excavate, grade, or otherwise alter the surface of the land so as to affect the capacity of a watercourse; destroy or significantly alter vegetation near a watercourse; or install or construct a new structure or improve or expand an existing structure within or across a watercourse without first obtaining a written permit from the Planning Director, and where the City may impose as conditions of approval any conditions reasonably related to the public health, safety, and welfare, including but not limited to riparian habitat restoration. Under Section 17.17.1 of the Municipal Code, when construction of a new residential structure is required to include a landscape plan, the plan must provide appropriate native riparian vegetation and other improvements designed to protect the environment (Section 18.04.010).

⁴ The City defines “watercourse” as any natural or machine-made channel or other facility or structure for directing and transporting water, whether continuously flowing or intermittent, improved or unimproved. Watercourse includes but is not limited to a streambed and the banks, drainage pipes and conduits, swales, ditches, washes, arroyos, dikes, and retaining walls associated with the streambed.

3.8 HYDROLOGY AND WATER QUALITY

Chapter 18.05 (Flood Hazard Reduction)

Chapter 18.05, Article 4 of the Municipal Code identifies specific requirements for development in areas that are within a flood hazard area as delineated by FEMA on the Flood Insurance Rate Maps.

3.8.3 IMPACTS AND MITIGATION MEASURES

STANDARDS OF SIGNIFICANCE

Based on Appendix G of the California Environmental Quality Act (CEQA) Guidelines, a hydrology or water quality impact is considered significant if project implementation would result in any of the following:

- 1) Violate any water quality standards or waste discharge requirements.
- 2) Substantially deplete groundwater supplies or interfere substantially with groundwater recharge such that there would be a net deficit in aquifer volume or a lowering of the local groundwater table level (e.g., the production rate of pre-existing nearby wells would drop to a level which would not support existing land uses or planned uses for which permits have been granted).
- 3) Substantially alter the existing drainage pattern of the site or area, including through the alteration of the course of a stream or river, in a manner which would result in substantial erosion or siltation on- or off-site.
- 4) Substantially alter the existing drainage pattern of the site or area, including through the alteration of the course of a stream or river, or substantially increase the rate or amount of surface runoff in a manner which would result in flooding on- or off-site.
- 5) Create or contribute runoff water which would exceed the capacity of existing or planned stormwater drainage systems or provide substantial additional sources of polluted runoff.
- 6) Otherwise substantially degrade water quality.
- 7) Place housing within a 100-year flood hazard area as mapped on a federal Flood Hazard Boundary or Flood Insurance Rate Map or other flood hazard delineation map.
- 8) Place within a 100-year flood hazard area structures which would impede or redirect flood flows.
- 9) Expose people or structures to a significant loss, injury or death involving flooding, including flooding as a result of the failure of a levee or dam.
- 10) Inundation by seiche, tsunami, or mudflow.

METHODOLOGY

The hydrology and water quality analysis presented in this section is based on a review of published information, regulatory agency data, technical reports, the City's General Plan, and environmental documents for projects in Orinda. Because a specific project that would implement one of the alternatives has not been designed, the analysis qualitatively evaluates the potential for the alternative sites to affect surface water and groundwater resources and water quality.

The impact analysis assumes that the design of future projects would be required to comply with applicable federal and state water quality and laws and regulations, which is monitored and enforced by the City through the Municipal Code. Chapter 18 of the Municipal Code contains specific requirements for development projects regarding the management of stormwater runoff, water quality, and development in a 100-year floodplain. Compliance with these regulations is required, not optional, and compliance must be demonstrated by the project applicant to have been incorporated in the project's design before permits are issued for project construction.

Impacts Not Further Evaluated

Flood Hazards

None of the alternative sites is in a FEMA-designated 100-year floodplain. There would be no impact for the alternative sites. General Plan Safety Element Policy 4.2.1.C directs that development must be located away from flood-prone areas unless flood risks can be mitigated. Any market-rate unit or second unit developed near a creek or stream for which there is a 100-year flood hazard would be required to comply with Chapter 18.05 (Flood Hazard Reduction) of the Municipal Code. Further, Section 18.03.050 of the Municipal Code prohibits impairing, impeding, redirecting, or otherwise affecting waters in a watercourse or otherwise altering the surface of the land in a manner that would affect the capacity of a watercourse. This would apply to market-rate units and detached second units to ensure no structures are built that could impede or redirect flood flows in a 100-year flood hazard area. The potential emergency shelter sites in Downtown are not within a 100-year flood hazard area. There would be no impact related to standards of significance 7 and 8, and these impacts do not require further evaluation.

The alternative sites are not located in an area that has been mapped by Contra Costa County as being at risk of dam failure inundation. There would be no impact for the alternative sites. For market-rate units and second units that could be developed in the inundation area for Briones Dam, the City would make a determination whether to permit development if there is potential for a parcel to be affected by flooding, consistent with General Plan Policy 4.2.1.C. The area in Downtown that could accommodate potential emergency shelters is not within a dam failure inundation area. There would be no impact related to standard of significance 9, and this impact does not require further evaluation.

The only lake in Orinda with a potential for seiche is Lake Cascade, a privately owned and maintained lake in the country club neighborhood. The alternative sites are not near the lake. There are no tsunami hazards in the City because of Orinda's inland location. There would be no impact relative to standard of significance 10 regarding seiche or tsunami, and this impact does not require further evaluation.

3.8 HYDROLOGY AND WATER QUALITY

None of the alternative sites is in an area where mudflow could pose a hazard. There would be no impact for the alternative sites relative to standard of significance 10. The slopes on the west side of Orinda have the highest susceptibility to mudflow. However, much of the west side is undeveloped EBMUD watershed land and open space. Steeper slopes are present on scattered locations zoned residential throughout the City that are also potential mudflow source areas. For those locations where a market-rate unit or second unit could be constructed, the City would make a determination, after consideration of General Plan Policy 4.2.1.C, whether to permit development if there is a potential for a parcel to be affected by mudflow. Project applicant(s) would be required to submit design-level, site-specific geotechnical reports and building plans to the City of Orinda for review and approval to address mudflow hazard, if present. This impact does not require further evaluation.

IMPACTS AND MITIGATION MEASURES

Water Quality (Standards of Significance 1, 3, 5, and 7)

Impact 3.8.1 The proposed project could generate stormwater runoff containing pollutants from construction sites and new impervious surfaces, which could affect water quality. This is a **less than significant** impact.

Impact Overview

Construction activities could temporarily disturb soils. Disturbed soils are susceptible to high rates of erosion from wind and rain, resulting in potential sediment transport from the project sites to nearby creeks and streams. Erosion and sedimentation could affect water quality, which could affect organisms and aquatic species in the watercourses. Other pollutants, such as nutrients, trace metals, and hydrocarbons, can attach to sediment and be transported with sediment to downstream locations. Sediment-associated pollutants could also cause or contribute to degradation of surface water quality. The delivery, handling, and storage of construction materials and wastes, as well as the use of construction equipment containing fuel, oil, and grease, could also introduce a risk for contamination that could impact surface water or groundwater quality as result of spills or leaks from heavy equipment and machinery. Equipment and material staging areas could be sources of surface water and groundwater pollution because of the use of paints, solvents, cleaning agents, and metals during construction. Pesticide use (including herbicides and fungicides) associated with site preparation work (as opposed to pesticide use for landscaping) is another potential source of contamination related to construction activities. Therefore, construction activities could impact surface water and groundwater quality in a manner that could lead to violations of water quality standards, if controls are not in place to minimize potential impacts.

The modification of a primarily vacant or underutilized area to an area with buildings with rooftops, parking, and landscaping would change the types and amounts of urban pollutants in stormwater runoff, compared to existing conditions because there would be more impervious surface over which stormwater would flow. Urban pollutants likely to be present in stormwater runoff could include oil, grease, and metals accumulated in roadways, driveways, parking lots, and on rooftops (from atmospheric dispersion), as well as pesticides, herbicides, particulate matter, nutrients, and animal waste from landscaped areas. Stormwater discharged to local creeks and streams through the City's drainage system or overland flow could cause receiving water quality degradation. This has the potential to violate water quality standards, if measures are not in place to minimize the types and amounts of pollutants in the runoff. Sedimentation and erosion would not be a substantial component of post-construction stormwater runoff because soils exposed during construction would be covered with impervious surfaces (buildings, parking areas, hardscaping) or landscaped.

Existing Standards That Address Impacts

General Plan Policy 4.1.1.E requires the City to protect creeks from siltation, pollution, and debris buildup. Policy 4.1.2.H requires the review of development proposals to ensure projects include controls to minimize runoff volumes and erosion. The City of Orinda is required to comply with all applicable federal and state laws and regulations pertaining to water quality protection. This includes laws and regulations under the Clean Water Act and the Porter-Cologne Water Quality Control Act. City ordinances expressly incorporate these standards into the Municipal Code.

Any project larger than 1 acre must comply with the NPDES Construction General Permit. The City is required to verify that the applicant has filed a Notice of Intent under the Construction General Permit and has prepared a SWPPP before issuing a grading or building permit. Projects smaller than 1 acre requiring a grading permit must comply with Orinda Municipal Code Chapters 18.02 and 18.03, which set forth specific requirements for stormwater runoff and erosion controls (BMPs) from construction sites. The controls such as BMPs required by the City provide an equivalent level of construction site runoff protection as required under the Construction General Permit.

Municipal Code Section 18.02.040.B requires that every application for a development project, including but not limited to a rezoning, tentative map, parcel map, conditional use permit, variance, site development permit, design review, or building permit that is subject to the development runoff requirements in the MRP, be accompanied by a stormwater control plan that meets the criteria in the most recent version of the Contra Costa Clean Water Program Stormwater C.3 Guidebook. Under Section 18.02.040.B, implementation of an approved stormwater control plan and submittal of an approved stormwater control operation and maintenance plan by the applicant must occur prior to issuance of a certificate of occupancy for a project subject to the requirements of Municipal Code Chapter 18.02.

Impacts Associated with Development of Alternative Housing Sites

Alternative 1

The Santa Maria Church site is generally undeveloped, consisting of a sports field and baseball diamond on maintained turfgrass and an asphalt basketball court in the southeastern portion of the site. The site does not contain any natural watercourses, and drainage is toward San Pablo Creek.

Site development at the Santa Maria Church site would involve the removal of pavement and vegetation, which would expose soil to erosion during construction. There would be heavy equipment and materials use, which could be a source of pollutants. Stormwater runoff from the site could enter San Pablo Creek. Because the site is greater than 1 acre, the project applicant's construction contractor would be required to implement a SWPPP and best management practices in accordance with the NPDES Construction General Permit. This would ensure development at the Alternative 1 site would not violate water quality standards, substantially increase erosion or sedimentation, or otherwise result in water quality pollution during construction.

After construction, Alternative 1 would result in most of the 3.2-acre site covered with impervious surfaces associated with the building rooftop, parking lot, sidewalks, and hardscape. There would be an increase in impervious surfaces (approximately 3 acres compared to existing conditions), which would increase the rate and amount of stormwater runoff from the site. Runoff could contain additional pollutant loads from the rooftops and parking lot and would differ in terms of the types of pollutants and the levels of pollutants compared to existing

3.8 HYDROLOGY AND WATER QUALITY

conditions. Prior to obtaining a grading permit and drainage permit from the City, the project applicant would need to identify the specific drainage controls to reduce stormwater runoff from the site and the specific water quality protection features that would be included in project design to demonstrate compliance with the Provision C.3 of the MRP, including LID requirements, in accordance with Municipal Code Section 18.02. This would ensure development at the Alternative 1 site would not contribute substantial pollutant loads to the storm drain system or otherwise degrade water quality.

Construction and operational impacts would be **less than significant**.

Alternative 2

The church sites are partially developed with parking lots. Drainage at the Orinda Community Church and St. Mark's Church sites is toward San Pablo Creek. Drainage from the St. John Orthodox Church site is to San Leandro Creek.

Site development at the three church locations would involve the removal of pavement and vegetation, which would expose soil to erosion during grading. There would be heavy equipment and materials use, which could be a source of pollutants. Stormwater runoff from the site could enter San Pablo Creek from the Orinda Community Church and the St. Mark's Church site, and could enter San Leandro Creek from the St. John Orthodox Church site. For the St. John Orthodox Church site, the project applicant's construction contractor would be required to implement a SWPPP and best management practices in accordance with the Construction General Permit because it is larger than 1 acre in size. For the Orinda Community Church and St. Mark's Church sites, which are less than 1 acre, the construction contractor would be required to implement construction site stormwater BMPs in accordance with Municipal Code Chapter 18. The BMPs required by the City provide an equivalent level of construction site runoff protection as required under the Construction General Permit. This would ensure development of Alternative 2 would not violate water quality standards, substantially increase erosion or sedimentation, or otherwise result in water quality pollution during construction.

After construction, the church sites would be covered with impervious surfaces associated with the building rooftop, parking lot, sidewalks, and hardscape. There would be an increase in impervious surfaces (approximately 1.5 acres total for the three sites compared to existing conditions; each individual site would be less than that), which would increase the rate and amount of stormwater runoff from the site. Runoff could contain additional pollutant loads from the rooftops and parking lot and would differ in terms of the types of pollutants and the levels of pollutants compared to existing conditions. Prior to obtaining a grading permit and drainage permit from the City, the project applicant would need to identify the specific drainage controls to reduce stormwater runoff from the site and the specific water quality protection features that would be included in project design to demonstrate compliance with the Provision C.3 of the MRP, including LID requirements, in accordance with Municipal Code Section 18.02. This would ensure development of Alternative 2 would not contribute substantial pollutant loads to the storm drain system or otherwise degrade water quality.

Construction and operational impacts would be **less than significant**.

Alternative 3

The 27 Orinda Way (Rite Aid) and 1 Orinda Way sites are completely covered with impervious surfaces. Construction would involve removal of pavement, which would expose soil to erosion during grading. There would be heavy equipment and materials use, which could be a source of pollutants. Stormwater runoff from the site could enter San Pablo Creek. For the 27 Orinda

Way (Rite Aid) site, the project applicant's construction contractor would be required to implement a SWPPP and best management practices in accordance with the Construction General Permit because it is larger than 1 acre in size. For the Orinda Community Church and 1 Orinda Way sites, which are less than 1 acre, the construction contractor would be required to implement construction site stormwater BMPs in accordance with Municipal Code Chapter 18. The BMPs required by the City provide an equivalent level of construction site runoff protection as required under the Construction General Permit. This would ensure development of Alternative 3 would not violate water quality standards, substantially increase erosion or sedimentation, or otherwise result in water quality pollution during construction.

After construction, the Orinda Community Church site would be covered with impervious surfaces associated with the building rooftop, parking lot, sidewalks, and hardscape. There would be an increase in impervious surfaces (approximately 0.4 acres compared to existing conditions), which would increase the rate and amount of stormwater runoff from that location. However, the 27 Orinda Way (Rite Aid) and 1 Orinda Way locations are already developed with 100 percent impervious surfaces with buildings and parking lots. There would be little, if any, change in stormwater runoff quality. The additional runoff from the church site could contain additional pollutant loads from the rooftops and parking lot and would differ in terms of the types of pollutants and the levels of pollutants compared to existing conditions. For all three locations, prior to obtaining a grading permit and drainage permit from the City, the project applicant would need to identify the specific drainage controls to reduce stormwater runoff from the sites and the specific water quality protection features that would be included in project design to demonstrate compliance with the Provision C.3 of the MRP, including LID requirements, in accordance with Municipal Code Section 18.02. This would ensure development of Alternative 3 would not contribute substantial pollutant loads to the storm drain system or otherwise degrade water quality.

Construction and operational impacts would be **less than significant**.

Impacts Associated with Development of Market-Rate Units, Second Units, Transitional/Supportive Housing, and Emergency Shelters

For the market-rate units that could be available in the Wilder subdivision, the potential environmental effects related to potential water quality degradation from construction and occupancy were addressed in the certified EIR for that project, as amended by the supplemental EIR and second supplemental EIR. The mitigation measures identified in the certified EIR, as amended, applied to site development and grading, which has been completed. This included stormwater quality BMPs as part of the storm drainage infrastructure serving the development. No new or more substantial construction or operational water quality impacts other than those evaluated and mitigated in the certified EIR are anticipated. For market-rate and second units and transitional/supportive housing constructed on vacant parcels elsewhere in the City, a building permit, grading permit, and possibly a drainage permit (depending on the parcel size) would be required. The project applicant would be required to have an erosion control plan approved by the City, which would identify how construction would minimize the potential for pollutants in stormwater runoff during construction. Stormwater runoff from individual residences and detached second units would not contribute substantial pollutant loads to the storm drain system. When the City reviews the permits and design review application, it will ensure the project complies with applicable standards for water quality protection. In addition, any residence constructed near a stream or creek would be required to include additional measures to protect water quality in accordance with Municipal Code Sections 18.02 and 17.17.1. Emergency shelters would not be a new source of stormwater runoff containing pollutants. Construction and operational impacts would be **less than significant**.

3.8 HYDROLOGY AND WATER QUALITY

Mitigation Measures

None required.

Groundwater (Standard of Significance 2)

Impact 3.8.2 The proposed project would not substantially deplete groundwater supplies or interfere substantially with groundwater recharge such that there would be a net deficit in aquifer volume or a lowering of the local groundwater table level. This impact is **less than significant**.

Impact Overview

Groundwater recharge potential can be affected if a project increases the amount of impervious surfaces in an important recharge area or if interferes with recharge potential of surface water. Changes in surface water volume and flow can be the result of alteration of a streambed or through diversions (surface water withdrawal). These changes can reduce the amount of groundwater available in an aquifer by reducing the volumes and/or lowering groundwater levels.

Existing Standards That Address Impacts

Section 18.03.050 of the Municipal Code prohibits impairing, impeding, redirecting, or otherwise affecting waters in a watercourse or otherwise altering the surface of the land in a manner that would affect the capacity of a watercourse.

Impacts Associated with Development of Alternative Sites

Groundwater recharge potential at the alternative sites is minimal. The individual sites are small, and redevelopment of the housing units would not result in a substantial change in the amount of impervious surface. None of the alternatives would result in the alteration of a stream channel because none of the sites are near any creeks or streams. Groundwater is not a source of supply to the potential housing sites, and no wells would be constructed. The alternative housing sites would be served with potable water through connection to the EBMUD water supply. Therefore, development of the alternative sites would not substantially deplete groundwater supplies or interfere substantially with groundwater recharge such that there would be a net deficit in aquifer volume or a lowering of the local groundwater table level. Impacts would be **less than significant**.

Impacts Associated with Development of Market-Rate Units, Second Units, Transitional/Supportive Housing, and Emergency Shelters

All market-rate units, second units, transitional/supportive housing, and emergency shelters would require connection to the EBMUD water system. Groundwater would not be used. Housing units would not be permitted along a creek or stream if the site development footprint would affect the capacity of the watercourse in a manner that would reduce the extent of alluvial soils along the streambed that may be available for recharge. Emergency shelters could only be developed in an area of Downtown where there is no surface water. There would be **no impact**.

Mitigation Measures

None required.

Drainage Patterns (Standards of Significance 3 and 4)

Impact 3.8.3 The proposed project would not substantially affect drainage patterns. The impact is **less than significant**.

Impact Overview

Drainage patterns of a site may be altered by grading, excavation, or cut-and-fill activities that alter the site's topography. Changes in drainage patterns could result in the redirection of stormwater flows over a site. These can be temporary during construction activities, when alteration of drainage patterns has the potential to cause or exacerbate erosion if soils are exposed to rainfall. Permanent changes in drainage patterns in combination with the addition of new impervious surfaces can increase the rate and volume of stormwater runoff, which can cause or exacerbate flooding.

Existing Standards That Address Impacts

General Plan Policy 4.2.2.E requires that land development be consistent with the carrying capacity of nearby streams and waterways, and Policy 4.1.2.E calls for drainage easements along creeks to protect riparian vegetation and buildings from flooding. Projects that result in a net increase of 500 square feet or more of impervious surface area must have a drainage permit (Municipal Code Section 18.03.050). The City requires that all permit applicants with projects that exceed 500 square feet of new impervious surfaces or that alter existing drainage systems route their runoff to landscaped areas and away from the street and creeks to the extent practicable. Potential drainage impacts that could cause erosion or siltation during construction would be controlled through implementation of a Construction General Permit and/or a City grading permit, depending on the size of the project.

Impacts Associated with Development of Alternative Sites

Alternative 1

Development of the Santa Maria Church site would require grading and potentially cut and fill to accommodate the building and parking area within the current topography, which includes a gentle upward slope toward the Orindawoods subdivision. The resulting earthwork would create a site that is relatively flat, similar to existing conditions. This would modify drainage patterns, but not in a manner that would worsen the potential for erosion or flooding at off-site properties. During construction, temporary changes in drainage patterns and potential erosion and siltation effects would be mitigated through compliance with the NPDES Construction General Permit, as explained in Impact 3.8.1.

Any project developed at the site would require a grading permit and a drainage permit from the City, and site design would be required to demonstrate compliance with Municipal Code requirements, which would include installation of appropriate drainage control features to ensure the stability of the areas upslope from the site to minimize potential erosion and siltation potential. The project would connect to the City's storm drain and, as such, would be required to include on-site features to minimize the amount of stormwater runoff discharged to the City's storm drain system. With implementation of City-required drainage design features, Alternative 1 would not cause or exacerbate erosion or siltation potential or flooding as a result of changes in drainage patterns. The impact would be **less than significant**.

3.8 HYDROLOGY AND WATER QUALITY

Alternative 2

Development of the Orinda Community Church site would require grading and potentially cut and fill to accommodate the building and parking area within the current topography, which includes a transition slope to the Orindawoods subdivision. The resulting earthwork would create a site that is relatively flat, compared to existing conditions. This would modify drainage patterns, but not in a manner that would worsen the potential for erosion or flooding at off-site properties. Site design would be required to demonstrate compliance with Municipal Code requirements, which would include installation of appropriate drainage control features to ensure the stability of the areas upslope from the Orinda Community Church site to minimize erosion and siltation potential. The St. Mark's Church and St. John Orthodox Church locations are flat, and changes in drainage patterns would be minimal. During construction, temporary changes in drainage patterns and potential erosion and siltation effects at the three church sites would be mitigated through compliance with the NPDES Construction General Permit for the St. John Orthodox Church site because it is greater than 1 acre and through BMPs required as part of the grading permit for the Orinda Community Church and St. Mark's Church sites,, as explained in Impact 3.8.1.

Development of the church locations would require a grading permit and a drainage permit from the City, and site design would be required to demonstrate compliance with Municipal Code requirements. The project would connect to the City's storm drain and, as such, would be required to include on-site features to minimize the amount of stormwater runoff discharged to the City's storm drain system. With implementation of City-required drainage design features, Alternative 2 would not cause or exacerbate erosion or siltation potential or flooding as a result of changes in drainage patterns. The impact would be **less than significant**.

Alternative 3

Potential impacts at the Orinda Community Church site would be mitigated as described above. The 27 Orinda Way (Rite Aid) and 1 Orinda Way locations are already developed with impervious surfaces (buildings and parking lots). While there may be a different configuration of the building footprint and parking lot if this alternative is implemented, this would result in little or no permanent effect on drainage patterns. Stormwater flows generated from the locations would continue to be conveyed to the City's storm drain system and would be designed to control stormwater on-site to minimize the potential for off-site flooding. During construction, temporary changes in drainage patterns and potential erosion and siltation effects would be mitigated through compliance with the NPDES Construction General Permit for the 27 Orinda Way (Rite Aid) site because it is greater than 1 acre and through BMPs that are required to be implemented as part of the grading permit for the Orinda Community Church and 1 Orinda Way sites, as explained in Impact 3.8.1.

Development of Alternative 3 would require a grading permit and a drainage permit from the City, and site design would be required to demonstrate compliance with Municipal Code requirements. The project would connect to the City's storm drain and, as such, would be required to include on-site features to minimize the amount of stormwater runoff discharged to the City's storm drain system. With implementation of City-required drainage design features, Alternative 3 would not cause or exacerbate erosion or siltation potential or flooding as a result of changes in drainage patterns. The impact would be **less than significant**.

Impacts Associated with Development of Market-Rate Units, Second Units, Transitional/Supportive Housing, and Emergency Shelters

For the market-rate units that could be available in the Wilder subdivision, the potential environmental effects related to potential changes in drainage patterns resulting from construction and occupancy were addressed in the certified EIR for that project, as amended by the supplemental EIR and second supplemental EIR. The mitigation measures identified in the certified EIR, as amended, applied to site development and grading, which has been completed. This included incorporation of design features to ensure potential changes in drainage patterns would not cause or exacerbate erosion or siltation or result in off-site flooding. No new or more substantial construction or operational drainage impacts other than those evaluated and mitigated in the certified EIR are anticipated. For market-rate units and detached second units constructed on vacant parcels elsewhere in the City, a building permit, grading permit, and a drainage permit (depending on the parcel size) would be required. In addition, any residence constructed near a stream or creek would be required to include appropriate measures to ensure it does not alter drainage patterns in a manner that would cause siltation in the watercourse or increase flood potential. Attached second units and emergency shelters would not alter drainage patterns because they would not involve grading. Construction and operational impacts would be **less than significant**.

Mitigation Measures

None required.

Storm Drain Capacity (Standard of Significance 5)

Impact 3.8.4 The proposed project could increase the amount of stormwater runoff into the City's storm drain system. This is a **less than significant** impact.

Impact Overview

As noted in Impact 3.8.1, a change in land use from a primarily vacant or underutilized area to an area with buildings with rooftops, parking, and landscaping would change the rate and flow of stormwater runoff, compared to existing conditions, because there would be more impervious surface over which stormwater would flow. Increased stormwater flows, when combined with existing flows to the storm drain system, may affect the capacity of the storm drain system if there is inadequate capacity. If the additional flows cannot be accommodated in the system, there is the potential for localized street flooding, particularly during large storm events.

Existing Standards That Address Impacts

The rate and volume of stormwater from a development project must be determined in accordance with Section 16.64.060 of the Municipal Code. Section 16.64.070 identifies capacity criteria for designing project drainage infrastructure. The storm drainage plan must be submitted to the City Public Works and Engineering Services for review and approval prior to the issuance of a grading permit.

Impacts Associated with Development of Alternative Sites

Alternative 1

Under Alternative 1, there would be an increase in impervious surface at this site, conservatively assumed to be approximately 3 acres compared to existing conditions. The project applicant would be required to estimate the rate and volume of flow in accordance Section 16.64.060 of

3.8 HYDROLOGY AND WATER QUALITY

the Municipal Code, and design the on-site storm drainage system and connections to the City's storm drain system in accordance with Section 16.64.070. The City has not identified any storm drain capacity deficiencies in the vicinity of the Santa Maria Church site. Because the contribution of stormwater flows from Alternative 1 would not exceed the capacity of existing facilities, it would not increase the risk for localized street flooding. The impact is **less than significant**.

Alternative 2

Alternative 2 would result in an increase in impervious surfaces at the three sites (approximately 1.5 acres combined; each individual site would be less than that). The project applicant(s) would be required to estimate the rate and volume of flow from each location in accordance Section 16.64.060 of the Municipal Code, and design the on-site storm drainage system and connections to the City's storm drain system for each location in accordance with Section 16.64.070. The City has not identified any storm drain capacity deficiencies in the vicinity of the Alternative 2 locations. Because the contribution of stormwater flows from Alternative 2 would not exceed the capacity of existing facilities, it would not increase the risk for localized street flooding. The impact is **less than significant**.

Alternative 3

Alternative 3 would result in an increase in impervious surfaces at the Orinda Community Church site (approximately 0.4 acres). The 27 Orinda Way (Rite Aid) and 1 Orinda Way locations are already developed with 100 percent impervious surfaces with buildings and parking lots. There would be little, if any, increase in runoff. The project applicant(s) would be required to estimate the rate and volume of flow from each location in accordance Section 16.64.060 of the Municipal Code, and design the on-site storm drainage system and connections to the City's storm drain system for each location in accordance with Section 16.64.070. The City has not identified any storm drain capacity deficiencies in the vicinity of the Alternative 3 locations. Because the contribution of stormwater flows from Alternative 3 would not exceed the capacity of existing facilities, it would not increase the risk for localized street flooding. The impact is **less than significant**.

Impacts Associated with Development of Market-Rate Units, Second Units, Transitional/Supportive Housing, and Emergency Shelters

For the market-rate units that could be available in the Wilder subdivision, the potential environmental effects on storm drainage system capacity were addressed in the certified EIR for that project, as amended by the supplemental EIR and second supplemental EIR. The mitigation measures identified in the certified EIR, as amended, applied to site development and grading, which has been completed. The necessary storm drain infrastructure and detention basin facilities have been constructed. No new or more substantial storm drain capacity impacts other than those evaluated and mitigated in the certified EIR are anticipated. For market-rate units and detached second units constructed on vacant parcels elsewhere, transitional/supportive housing and emergency shelters, site plans must include appropriate drainage controls to minimize runoff. Impacts would be **less than significant**.

Mitigation Measures

None required.

3.8.4 CUMULATIVE SETTING, IMPACTS, AND MITIGATION MEASURES

CUMULATIVE SETTING

The cumulative setting for water quality and drainage impacts is development in the San Pablo Creek and San Leandro Creek watersheds. The cumulative setting for storm drain system capacity impacts is buildout of the Orinda General Plan because the system only accommodates flows from the City.

CUMULATIVE IMPACTS AND MITIGATION MEASURES

Cumulative Impacts on Water Quality and Drainage Patterns

Impact 3.8.5 The proposed project, in combination with implementation of cumulative projects in Orinda, Moraga, and Lafayette within the San Pablo Creek and San Leandro Creek watersheds, could result in changes in drainage patterns that could alter stormwater flows and urban pollutants in runoff, which could affect water quality. The proposed project's contribution would be **less than cumulatively considerable**, and the cumulative impact is **less than significant**.

Cumulative development would result in changes in land use and associated construction and occupancy of projects that would generate additional stormwater flows and urban pollutants. Jurisdictions within the San Pablo Creek and San Leandro Creek watersheds are required to comply with all applicable federal and state water quality protection laws and regulations. Construction projects greater than 1 acre must comply with the Construction General Permit. For smaller projects, each jurisdiction ensures compliance with water quality standards through general plan policies, municipal code regulations, standards, and/or guidelines. Contra Costa County and its 19 incorporated cities are regulated waste dischargers under a Municipal Regional Stormwater NPDES Permit. Provision C.3 of the Municipal Regional Permit for New Development and Redevelopment allows permittees to use their planning authorities to include appropriate source control, site design, and stormwater treatment measures in new development and redevelopment projects to address both soluble and insoluble stormwater runoff pollutant discharges and prevent increases in runoff flows from new development and redevelopment projects. The incorporation of the LID requirements into new development and redevelopment would continue to reduce urban pollutants in the watershed to the maximum extent practicable.

Each of the three alternatives would involve no more than 3.2 acres of ground disturbance and development of impervious surfaces for the largest single site (Santa Maria Church, Alternative 1, and the three church sites combined [Alternative 2]) to less than 0.8 acre at the Orinda Community Church site (Alternatives 2 and 3). Due to the relatively small acreage, building footprint, and parking lots, each alternative would not be a substantial source of runoff containing pollutants, nor would it alter drainage patterns in a manner that would generate substantial erosion or siltation. None of the potential housing sites is adjacent to a creek or stream where runoff from a site would directly enter the watercourse. Under Municipal Code Chapter 18, the City would require site-specific drainage plans for any project, which must include stormwater best management practices that reflect LID principles. For these reasons, the proposed project's contribution to watershed-level water quality and drainage impacts would be **less than cumulatively considerable**, and the cumulative impact is **less than significant**.

Mitigation Measures

None required.

Cumulative Impacts on Storm Drainage Capacity

Impact 3.8.6 The proposed project, in combination with buildout of the General Plan, would not contribute flows to the City's storm drain system that would exceed capacity. The proposed project's contribution would be **less than cumulatively considerable**, and the cumulative impact is **less than significant**.

There are no major storm drain system deficiencies in the City resulting from the amount of runoff entering the system. Capacity issues are generally related to the age and condition of the storm drain lines themselves (as noted above) and not the rate or volume of flow from development. The City ensures storm drain capacity is sufficient to accommodate new flows in addition to existing flows by requiring development projects to include appropriate on-site controls and designing inlets and pipe sizing in accordance with City standards (Wellbrock 2014).

Each of the three alternatives would involve no more than approximately 3 acres increase in impervious surfaces for the largest single site (Santa Maria Church, Alternative 1, and the three church sites combined [Alternative 2]) to less than 0.8 acre at the Orinda Community Church site (Alternatives 2 and 3). Due to the relatively small size of each development location, none of the alternatives would contribute a substantial amount of runoff. Further, the selected alternative would be required to comply with City standards for on-site storm drain system sizing and design, as would be required for any project in the City meeting the criteria under Chapter 16.64 of the Municipal Code pertaining to storm drain systems. The proposed project's contribution would be **less than cumulatively considerable**, and the cumulative impact is **less than significant**.

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3.8 HYDROLOGY AND WATER QUALITY

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